EXPLORATORY DATA ANALYSIS USING POWER BI

WEEK 9 - HOUSING DATA CLEANING & TRANSFORMATION

REPORT

SUBMITTED BY

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1. Introduction

This practical focuses on cleaning and transforming the King County House Sales dataset using Power Query in Microsoft Power BI. The dataset contains historical house-sale records and property characteristics. Data cleaning and transformation help improve data quality and prepare the dataset for further analysis.

2. Problem Statement

A real-estate organization wants to analyze historical house sales data to understand property characteristics and the factors associated with house prices. The dataset must be cleaned, transformed, and organized to support accurate analysis of house prices and property features.

3. Objectives

- Import the housing dataset into Power Query.
- Identify and handle missing, invalid, and unsuitable data.
- Extract sale year and sale month from sale dates.
- Calculate property age at the time of sale.
- Calculate price per square foot and total area.
- Identify whether a property has been renovated.
- Categorize properties into size buckets and price tiers.
- Load the cleaned dataset into Power BI for further analysis.

4. Dataset and Tools Used

Dataset: “kc_ House_ Data.csv”

Software: Microsoft Power BI Desktop

Tool: Power Query Editor

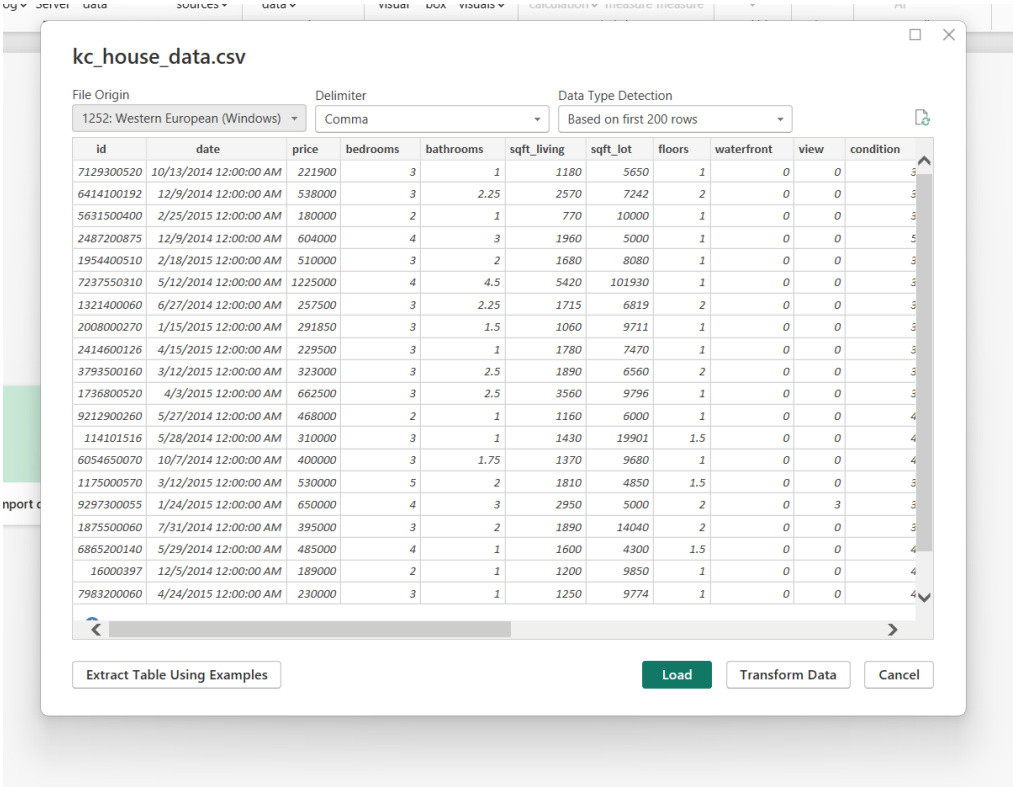
The dataset contains information about house prices, sale dates, bedrooms, bathrooms, living area, lot area, construction year, and renovation year.

5. Procedure

Step 1: Import the Dataset

- ✓ Open Microsoft Power BI Desktop.
- ✓ Click Home → Get Data → Text/CSV.
- ✓ Select the King County House Sales CSV file.
- ✓ Preview the dataset.
- ✓ Click Transform Data to open Power Query Editor.

The dataset is now ready for cleaning and transformation.



id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition
7129300520	10/13/2014 12:00:00 AM	221900	3	1	1180	5650	1	0	0	3
6414100192	12/9/2014 12:00:00 AM	538000	3	2.25	2570	7242	2	0	0	3
5631500400	2/25/2015 12:00:00 AM	180000	2	1	770	10000	1	0	0	3
2487200875	12/9/2014 12:00:00 AM	604000	4	3	1960	5000	1	0	0	5
1954400510	2/18/2015 12:00:00 AM	510000	3	2	1680	8080	1	0	0	3
7237550310	5/12/2014 12:00:00 AM	1225000	4	4.5	5420	101930	1	0	0	3
1321400060	6/27/2014 12:00:00 AM	257500	3	2.25	1715	6819	2	0	0	3
2008000270	1/15/2015 12:00:00 AM	291850	3	1.5	1060	9711	1	0	0	3
2414600126	4/15/2015 12:00:00 AM	229500	3	1	1780	7470	1	0	0	3
3793500160	3/12/2015 12:00:00 AM	323000	3	2.5	1890	6560	2	0	0	3
1736800520	4/3/2015 12:00:00 AM	662500	3	2.5	3560	9796	1	0	0	3
9212900260	5/27/2014 12:00:00 AM	468000	2	1	1160	6000	1	0	0	4
114101516	5/28/2014 12:00:00 AM	310000	3	1	1430	19901	1.5	0	0	4
6054650070	10/7/2014 12:00:00 AM	400000	3	1.75	1370	9680	1	0	0	4
1175000570	3/12/2015 12:00:00 AM	530000	5	2	1810	4850	1.5	0	0	3
9297300055	1/24/2015 12:00:00 AM	650000	4	3	2950	5000	2	0	3	3
1875500060	7/31/2014 12:00:00 AM	395000	3	2	1890	14040	2	0	0	3
6865200140	5/29/2014 12:00:00 AM	485000	4	1	1600	4300	1.5	0	0	4
16000397	12/5/2014 12:00:00 AM	189000	2	1	1200	9850	1	0	0	4
7983200060	4/24/2015 12:00:00 AM	230000	3	1	1250	9774	1	0	0	4

Step 2: Check Data Types

Check the data types of all columns and correct them where necessary.

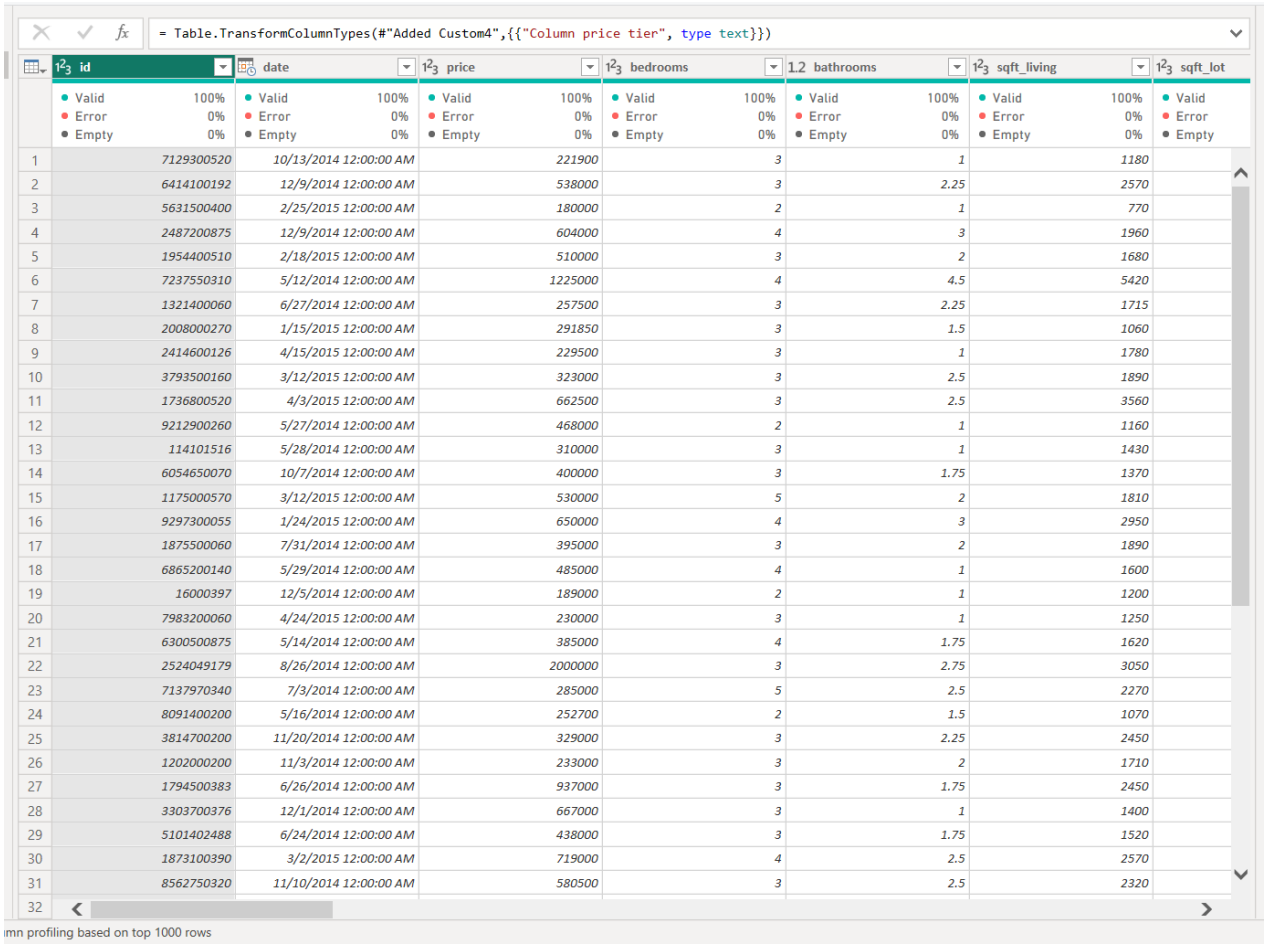
- ✓ Date: Date
- ✓ Price: Decimal Number or Whole Number
- ✓ Bedrooms: Whole Number
- ✓ Bathrooms: Decimal Number
- ✓ Living Area: Whole Number
- ✓ Lot Area: Whole Number
- ✓ Year Built: Whole Number
- ✓ Year Renovated: Whole Number.

Step 3: Clean the Dataset

- ✓ Check the columns for missing values, nulls, and errors.
- ✓ Remove duplicate records only when they are genuine duplicates.
- ✓ Check the bedroom and bathroom columns for zero or invalid values.
- ✓ Filter out infeasible records as required by the assignment.

- ✓ Check the price and area columns for invalid values.

This process improves the quality and reliability of the dataset.



	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot
1	7129300520	10/13/2014 12:00:00 AM	221900	3	1	1180	
2	6414100192	12/9/2014 12:00:00 AM	538000	3	2.25	2570	
3	5631500400	2/25/2015 12:00:00 AM	180000	2	1	770	
4	2487200875	12/9/2014 12:00:00 AM	604000	4	3	1960	
5	1954400510	2/18/2015 12:00:00 AM	510000	3	2	1680	
6	7237550310	5/12/2014 12:00:00 AM	1225000	4	4.5	5420	
7	1321400060	6/27/2014 12:00:00 AM	257500	3	2.25	1715	
8	2008000270	1/15/2015 12:00:00 AM	291850	3	1.5	1060	
9	2414600126	4/15/2015 12:00:00 AM	229500	3	1	1780	
10	3793500160	3/12/2015 12:00:00 AM	323000	3	2.5	1890	
11	1736800520	4/3/2015 12:00:00 AM	662500	3	2.5	3560	
12	9212900260	5/27/2014 12:00:00 AM	468000	2	1	1160	
13	114101516	5/28/2014 12:00:00 AM	310000	3	1	1430	
14	6054650070	10/7/2014 12:00:00 AM	400000	3	1.75	1370	
15	1175000570	3/12/2015 12:00:00 AM	530000	5	2	1810	
16	9297300055	1/24/2015 12:00:00 AM	650000	4	3	2950	
17	1875500060	7/31/2014 12:00:00 AM	395000	3	2	1890	
18	6865200140	5/29/2014 12:00:00 AM	485000	4	1	1600	
19	16000397	12/5/2014 12:00:00 AM	189000	2	1	1200	
20	7983200060	4/24/2015 12:00:00 AM	230000	3	1	1250	
21	6300500875	5/14/2014 12:00:00 AM	385000	4	1.75	1620	
22	2524049179	8/26/2014 12:00:00 AM	2000000	3	2.75	3050	
23	7137970340	7/3/2014 12:00:00 AM	285000	5	2.5	2270	
24	8091400200	5/16/2014 12:00:00 AM	252700	2	1.5	1070	
25	3814700200	11/20/2014 12:00:00 AM	329000	3	2.25	2450	
26	1202000200	11/3/2014 12:00:00 AM	233000	3	2	1710	
27	1794500383	6/26/2014 12:00:00 AM	937000	3	1.75	2450	
28	3303700376	12/1/2014 12:00:00 AM	667000	3	1	1400	
29	5101402488	6/24/2014 12:00:00 AM	438000	3	1.75	1520	
30	1873100390	3/2/2015 12:00:00 AM	719000	4	2.5	2570	
31	8562750320	11/10/2014 12:00:00 AM	580500	3	2.5	2320	
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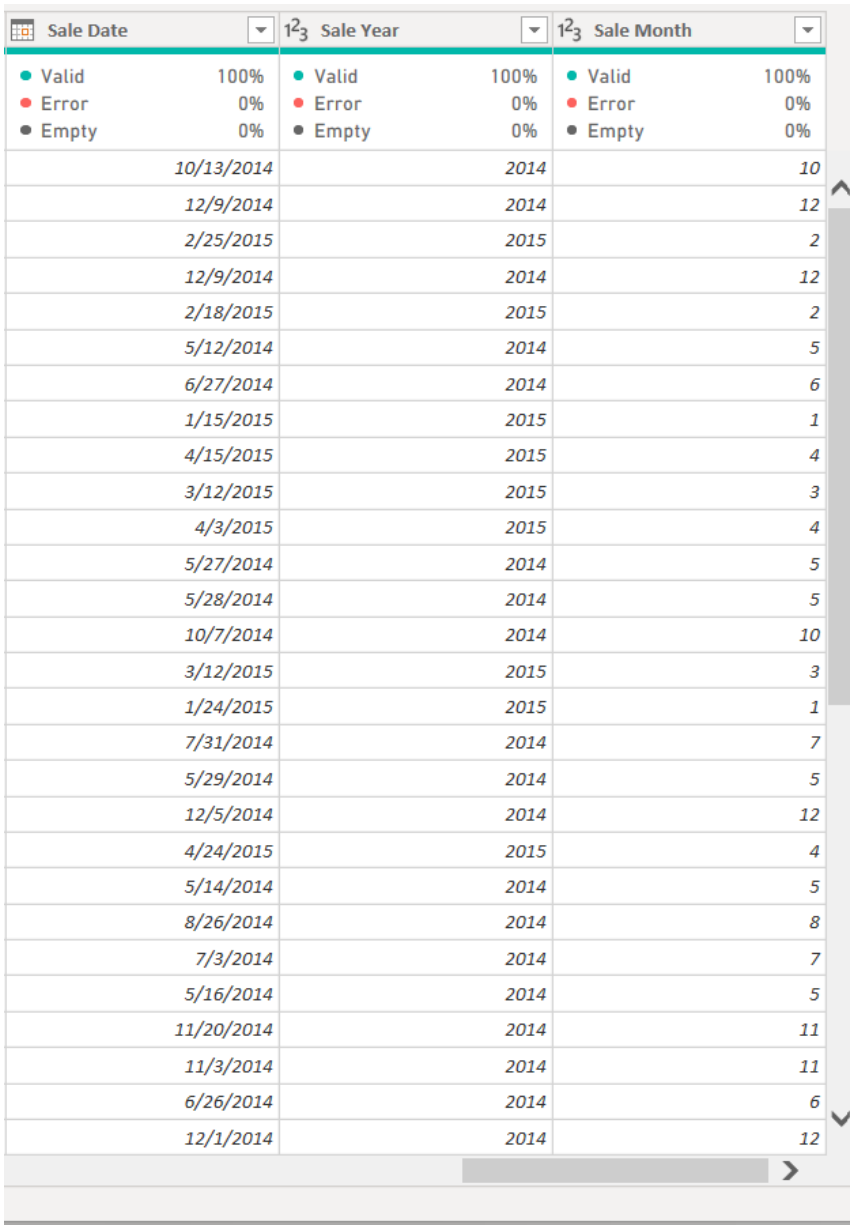
Step 4: Extract Date, Year and Month Columns

- ✓ Select the date column in Power Query Editor.
- ✓ Go to the Add Column tab.
- ✓ Select Date → Date Only to create a new column containing only the date.
- ✓ Again, select the original date column.
- ✓ Go to Add Column → Date → Year → Year to extract the sale year.
- ✓ Select the date column once again.
- ✓ Go to Add Column → Date → Month → Month to extract the sale month.

The following columns are created:

- Date Only
- Year
- Month

These columns help organize the housing data based on the date of sale and support time-based analysis.



Sale Date	Sale Year	Sale Month
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
10/13/2014	2014	10
12/9/2014	2014	12
2/25/2015	2015	2
12/9/2014	2014	12
2/18/2015	2015	2
5/12/2014	2014	5
6/27/2014	2014	6
1/15/2015	2015	1
4/15/2015	2015	4
3/12/2015	2015	3
4/3/2015	2015	4
5/27/2014	2014	5
5/28/2014	2014	5
10/7/2014	2014	10
3/12/2015	2015	3
1/24/2015	2015	1
7/31/2014	2014	7
5/29/2014	2014	5
12/5/2014	2014	12
4/24/2015	2015	4
5/14/2014	2014	5
8/26/2014	2014	8
7/3/2014	2014	7
5/16/2014	2014	5
11/20/2014	2014	11
11/3/2014	2014	11
6/26/2014	2014	6
12/1/2014	2014	12

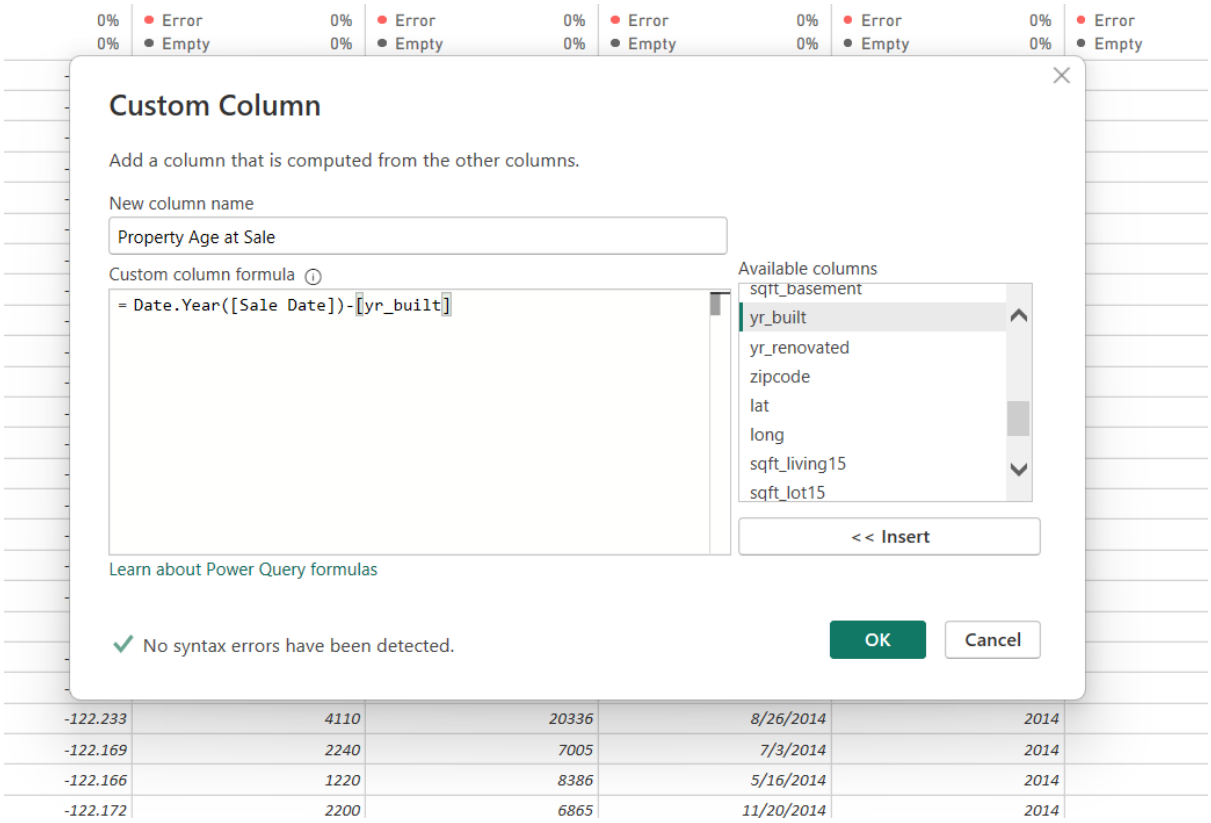
Step 5: Calculate Property Age at Sale

- ✓ Select Add Column → Custom Column.
- ✓ Enter the column name as Property Age at Sale.
- ✓ Enter the following formula:

  Date.Year([date]) - [yr_built]

- ✓ Click OK.

This calculates the approximate age of the property at the time of sale by subtracting the construction year from the sale year.



Step 6: Calculate Price per Square Foot

- ✓ Go to Add Column → Custom Column.
- ✓ Enter the new column name as Price per SqFt.

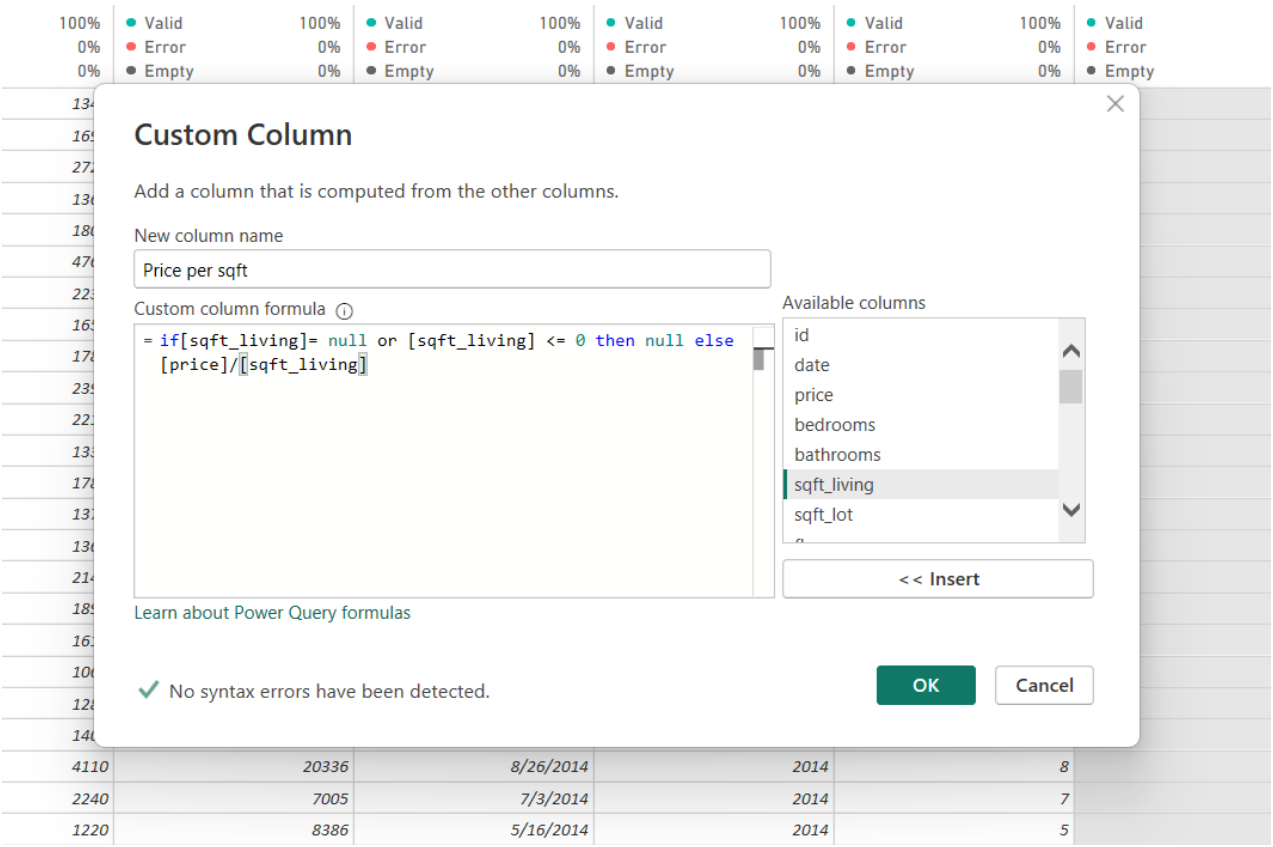
✓ Enter the following formula:

if [sqft_living] = null or [sqft_living] <= 0

then null else[price] / [sqft_living]

✓ Click OK.

This calculates the price per square feet using the house price and living area. The condition prevents division by zero or invalid living-area values.



Step 7: Calculate Total Area

✓ Go to Add Column → Custom Column.

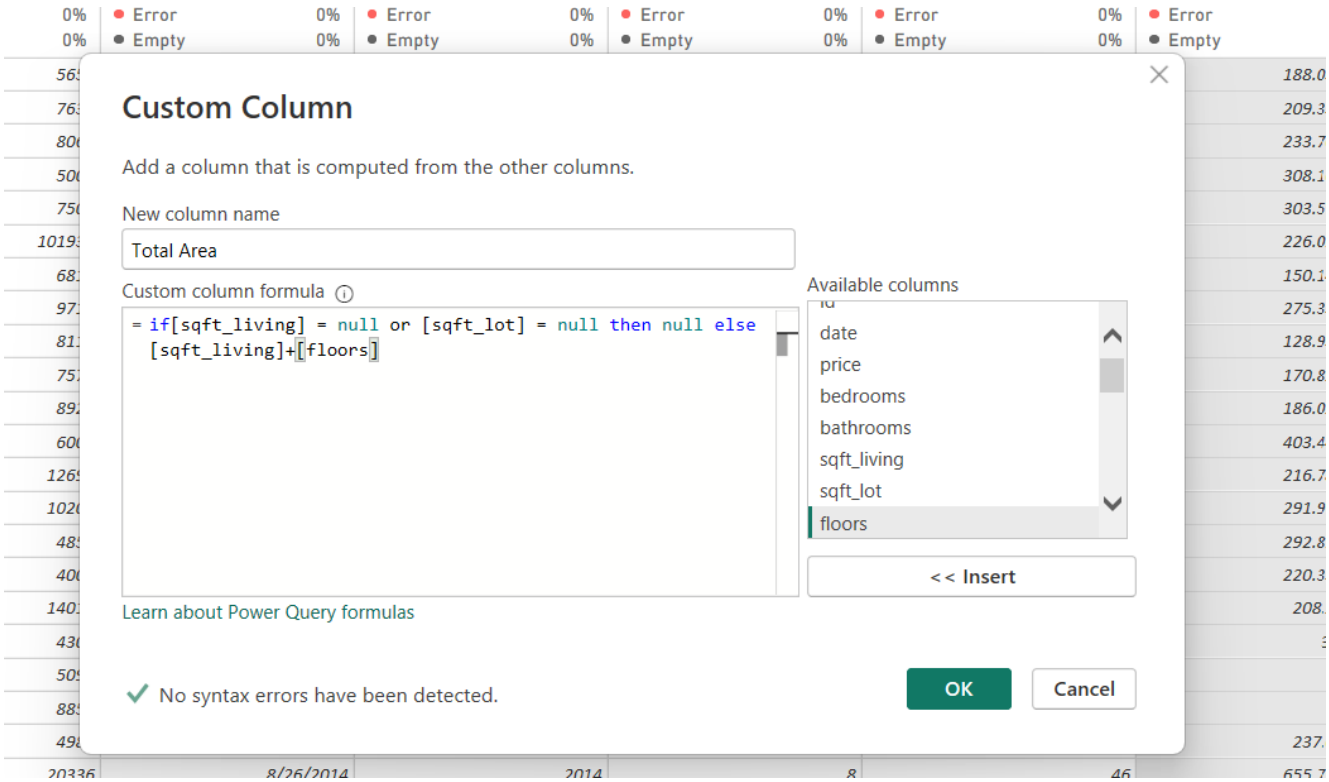
✓ Enter the new column name as Total Area.

✓ Enter the following formula:

if [sqft_living] = null or [sqft_lot] = null then null else[sqft_living] + [floors]

✓ Click OK.

This calculates the combined value of living area and lot area. It helps in understanding the overall property measurements.



Step 8: Create the Renovation Status Column

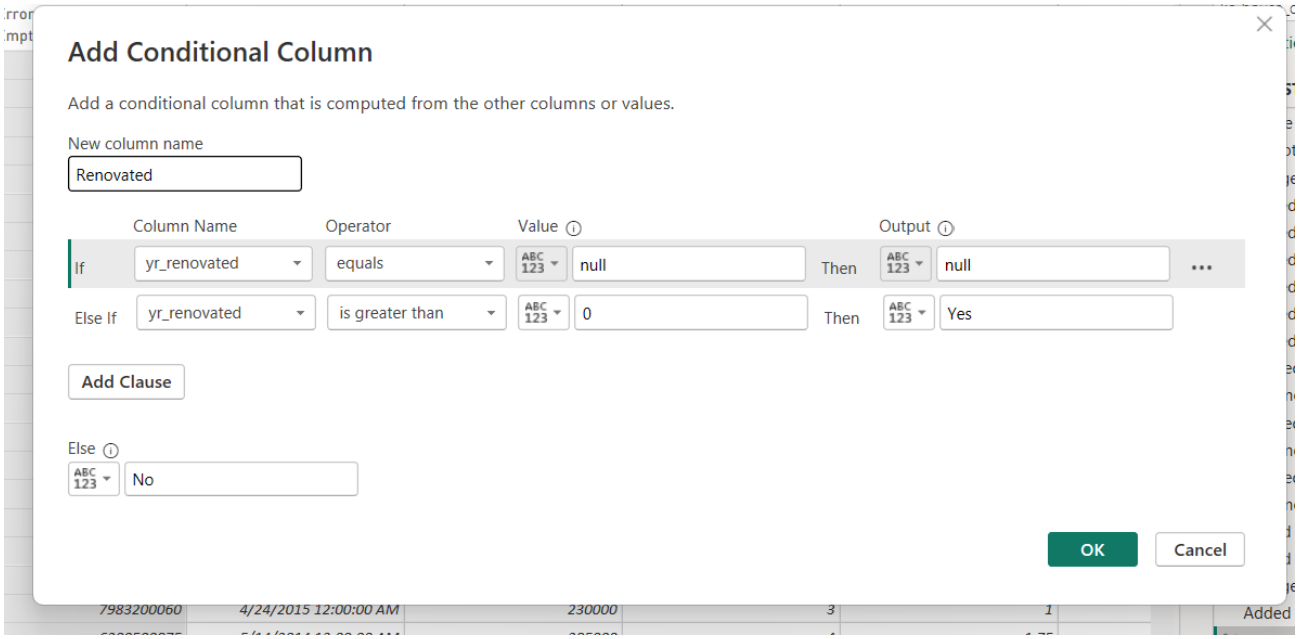
- ✓ Go to Add Column → Conditional Column.
- ✓ Enter the new column name as Renovated.
- ✓ Set the following condition:

Column Name	Operator	Value	Output
If yr_renovated	equals	null	null
Else if yr_renovated	is greater than	0	Yes
Else			No

- ✓ Click OK.

This identifies whether a property has been renovated.

- Yes: The property has a recorded renovation year.
- No: The property has no recorded renovation year.

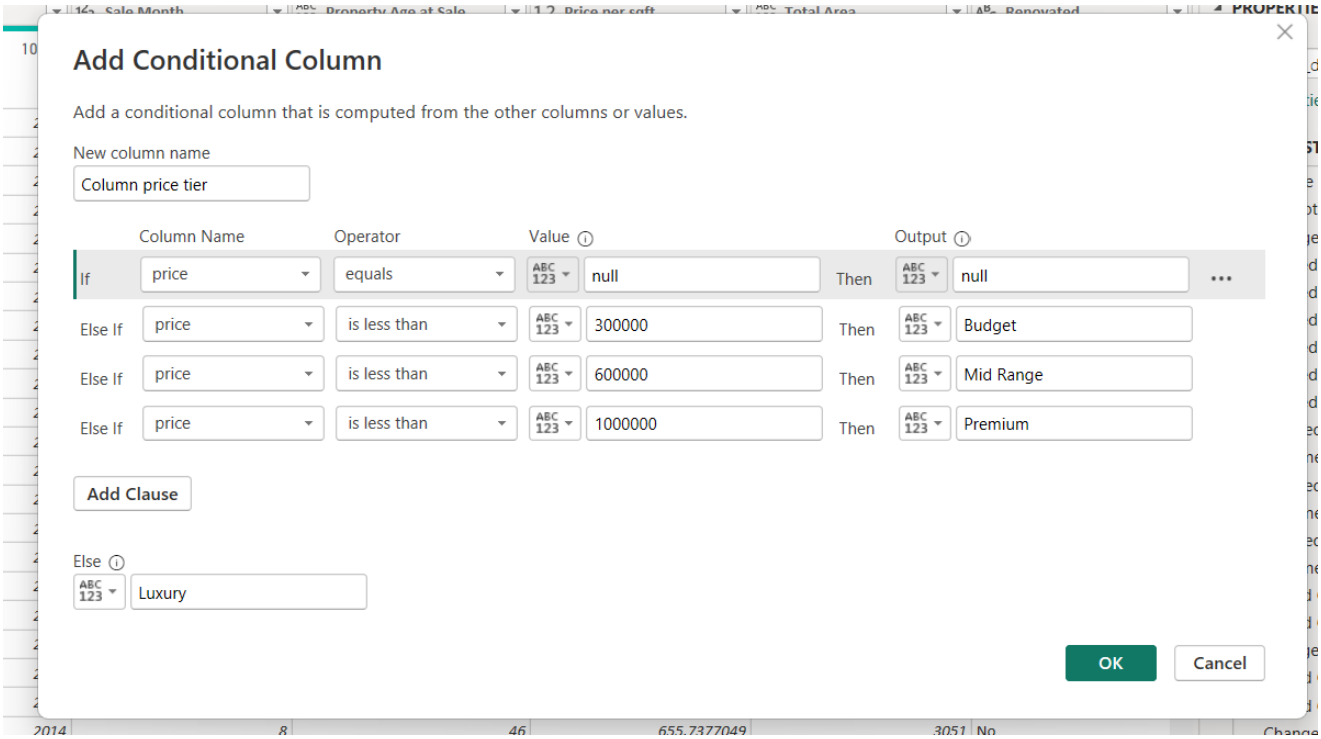


Step 9: Categorize Properties by Price

- ✓ Go to Add Column → Conditional Column.
- ✓ Enter the new column name as Column Price Tier.
- ✓ Configure the following conditions:

Column	Operator	Value	Output
price	equals	null	null
price	is less than	300000	Budget
price	is less than	600000	Mid-range
price	is less than	1000000	Premium
Else			Luxury

✓ Click OK.



Step 10: Categorize Properties by Size

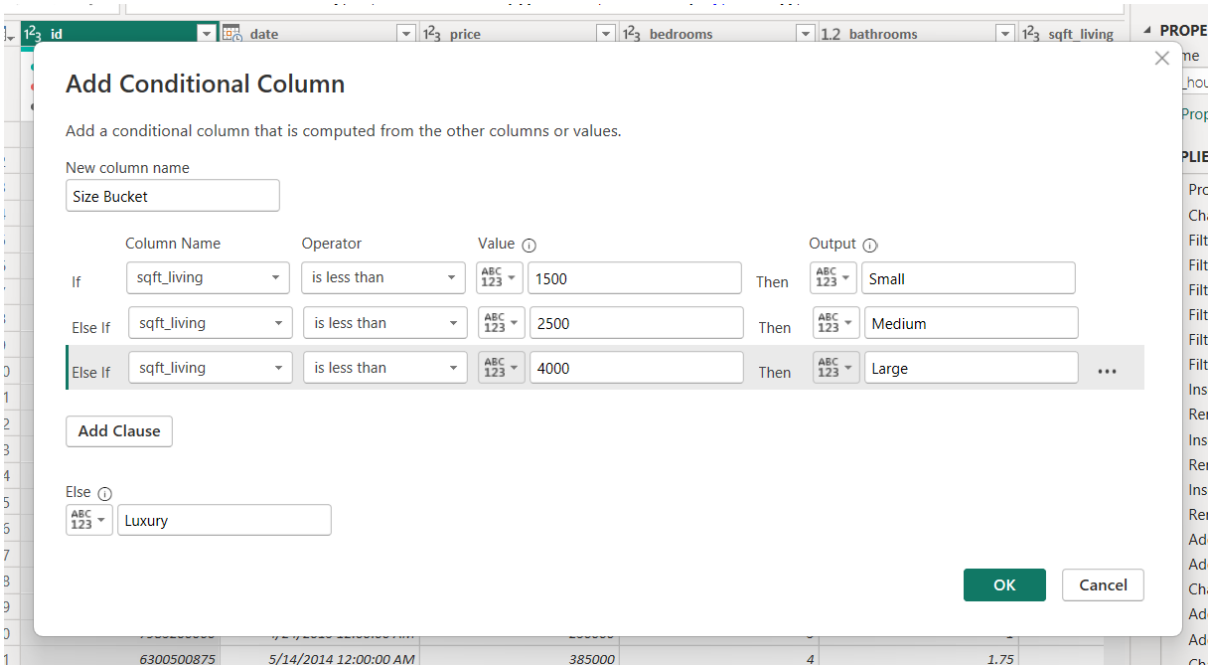
✓ Go to Add Column → Conditional Column.

✓ Enter the new column name as Size Bucket.

✓ Configure the following conditions:

Condition	Output
sqft_living is less than 1500	Small
sqft_living is less than 2500	Medium
sqft_living is less than 4000	Large
Else	Luxury

✓ Click OK.

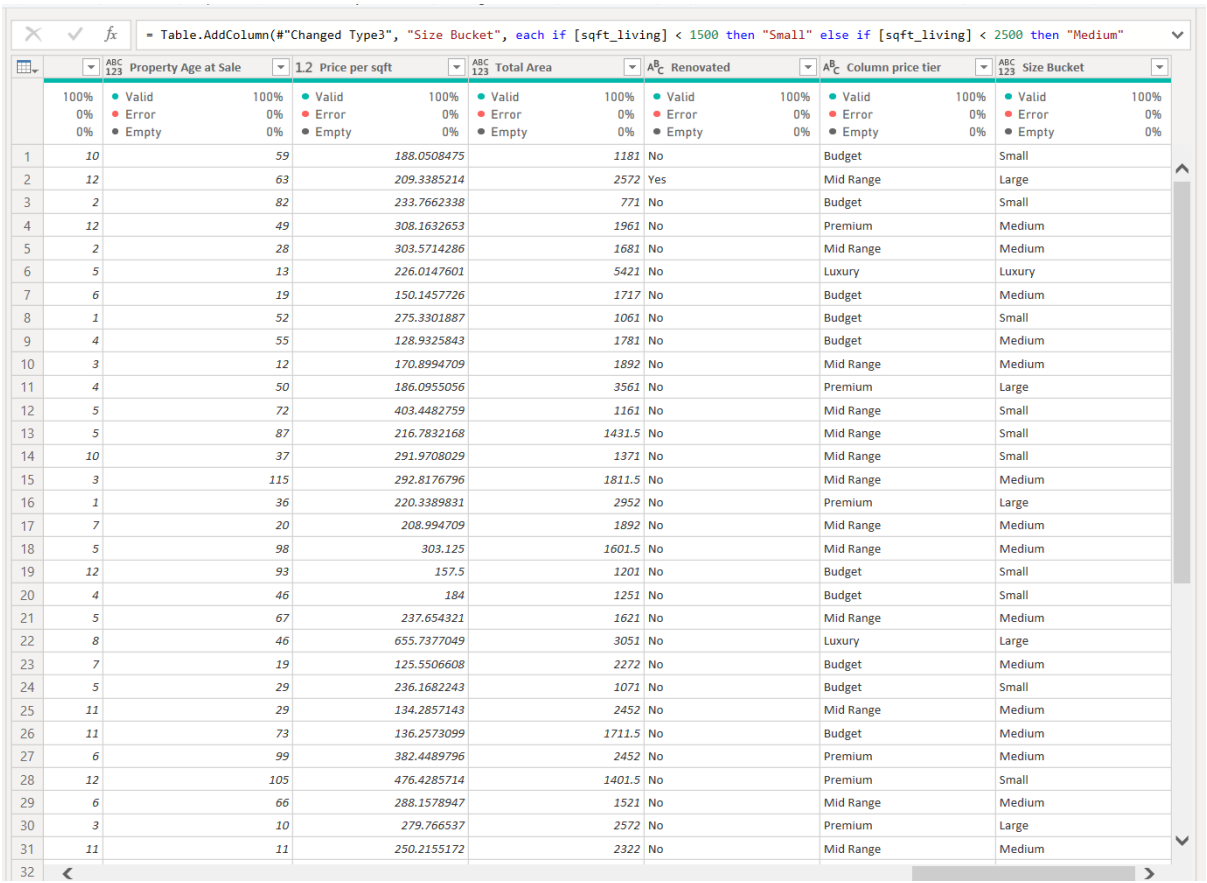


Outcome of all the six columns:

After completing the transformation process, the dataset contains cleaned house-sale records and additional columns, including:

- Property Age at Sale
- Price per SqFt
- Total Area
- Renovated
- Size Bucket
- Column Price Tier

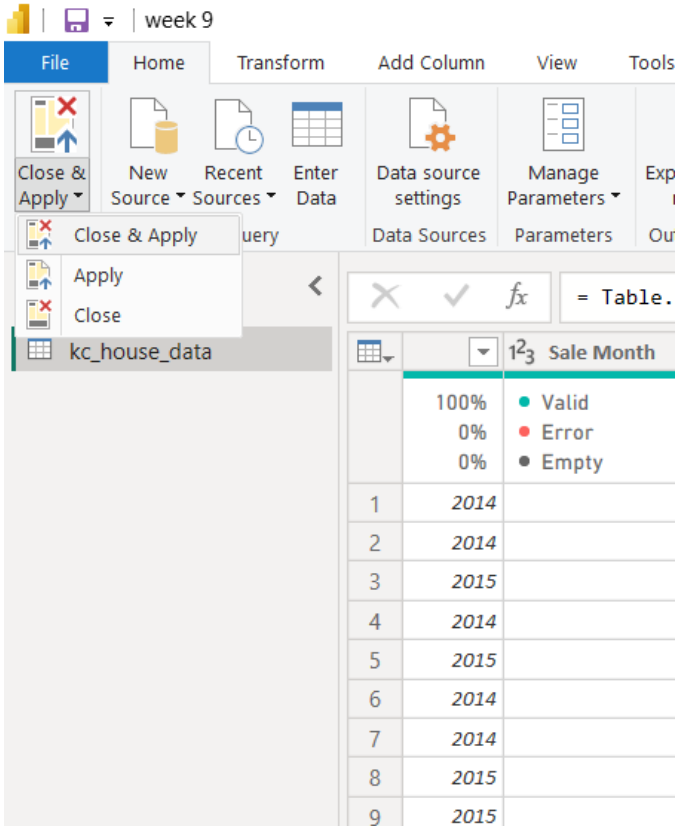
These columns make it easier to analyze housing data and understand relationships between property characteristics and sale prices.



```
= Table.AddColumn(#"Changed Type3", "Size Bucket", each if [sqft_living] < 1500 then "Small" else if [sqft_living] < 2500 then "Medium"
```

	Property Age at Sale	Price per sqft	Total Area	Renovated	Column price tier	Size Bucket
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1	10	59	188.0508475	1181	No	Budget	Small
2	12	63	209.3385214	2572	Yes	Mid Range	Large
3	2	82	233.7662338	771	No	Budget	Small
4	12	49	308.1632653	1961	No	Premium	Medium
5	2	28	303.5714286	1681	No	Mid Range	Medium
6	5	13	226.0147601	5421	No	Luxury	Luxury
7	6	19	150.1457726	1717	No	Budget	Medium
8	1	52	275.3301887	1061	No	Budget	Small
9	4	55	128.9325843	1781	No	Budget	Medium
10	3	12	170.8994709	1892	No	Mid Range	Medium
11	4	50	186.0955056	3561	No	Premium	Large
12	5	72	403.4482759	1161	No	Mid Range	Small
13	5	87	216.7832168	1431.5	No	Mid Range	Small
14	10	37	291.9708029	1371	No	Mid Range	Small
15	3	115	292.8176796	1811.5	No	Mid Range	Medium
16	1	36	220.3389831	2952	No	Premium	Large
17	7	20	208.994709	1892	No	Mid Range	Medium
18	5	98	303.125	1601.5	No	Mid Range	Medium
19	12	93	157.5	1201	No	Budget	Small
20	4	46	184	1251	No	Budget	Small
21	5	67	237.654321	1621	No	Mid Range	Medium
22	8	46	655.7377049	3051	No	Luxury	Large
23	7	19	125.5506608	2272	No	Budget	Medium
24	5	29	236.1682243	1071	No	Budget	Small
25	11	29	134.2857143	2452	No	Mid Range	Medium
26	11	73	136.2573099	1711.5	No	Budget	Medium
27	6	99	382.4489796	2452	No	Premium	Medium
28	12	105	476.4285714	1401.5	No	Premium	Small
29	6	66	288.1578947	1521	No	Mid Range	Medium
30	3	10	279.766537	2572	No	Premium	Large
31	11	11	250.2155172	2322	No	Mid Range	Medium
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Then, close and apply.



6. Conclusion

The Week 9 task demonstrates how Microsoft Power BI and Power Query can be used to clean and transform housing data. The dataset is prepared by checking data types, handling invalid records, extracting date-related information, calculating property age, and creating additional columns using conditional transformations.

The final transformed dataset provides a structured foundation for analyzing historical house sales, comparing property characteristics, and creating meaningful visualizations in Power BI.